

Marker:1b Armor Stands render themselves and their equipment dark if inside solid blocks

MC-107529 Lighting

Status	Resolved / Fixed
Affected	Minecraft 1.10.2, Minecraft 16w38a, Minecraft 16w39a, Minecraft 16w39b, Minecraft 16w39c, Minecraft 16w40a (+95 more)
Fixed in	1.16.2 Pre-release 3
Reported	2016-09-28

STEPS TO REPRODUCE

1. Create or open a Creative world.
2. Place a command block.
3. Place three solid blocks, such as smoothstone, on top of the command block.
4. Enter the following command into the command block: `/summon armor_stand ~ ~2 ~ {CustomNameVisible:1b,Glowing:1b,ShowArms:1b,Marker:1b,Invisible:0b,NoBasePlate:0b,HandItems:[{id:"minecraft:lapis_block",Count:1b},{id:"minecraft:lapis_ore",Count:1b}],ArmorItems:[{id:"minecraft:iron_boots",Count:1b},{id:"minecraft:iron_leggings",Count:1b},{id:"minecraft:chainmail_chestplate",Count:1b},{id:"minecraft:player_head",Count:1b,tag:{SkullOwner:{Id:"994924dc-8404-4af4-a58b-678426ee0095",Properties:{textures:[{Value:"eyJ0ZXh0dXJlcyl6eyJTS0lOljp7InVyYyYhYm90dHA6Ly90ZXh0dXJlcj5taW5lY3JhZnQubmV0L3RleHR1cmUvYzBmZmE0OWIxYWQ3ZmNmZDZhNWUxYTU4YzMyZjFkM2I5OTYwMDZjMTAzODBhNTQ0MTlkZmFmM2ZiNzc4MzdjZiJ9fX0="}}]}},CustomName:"{\\\"text\\\":\\\"Meri\\\",\\\"color\\\":\\\"dark_blue\\\",\\\"bold\\\":\\\"true\\\",\\\"italic\\\":\\\"true\\\"}"}`
5. Trigger the command block to summon the armor stand.
6. Run the following command repeatedly to move the armor stand downward in increments of 0.1 blocks: `/execute as @e[type=armor_stand] at @s run tp @s ~~-0.1 ~``
7. Observe that, once part of the armor stand is inside a solid block, the armor stand and its equipment render dark.

OBSERVED BEHAVIOUR

Actual behavior

When the `Marker:1b` armor stand is gradually teleported downward into the three-block-high smooth stone column, its uppermost hitbox portion eventually intersects a solid block. At that point, the armor stand model and all of its equipped items render very dark or black, even though the armor stand has the `Marker` tag and therefore uses the reduced marker hitbox.

The dark rendering persists while the relevant portion of the armor stand is inside the solid block. The effect also applies to equipment being rendered by the armor stand, including its armor, held items, and player head. Using transparent blocks instead prevents the issue, but the armor stand cannot render normally while intersecting a solid block.

EXPECTED BEHAVIOUR

A Marker:1 armor stand should continue rendering itself, its name, and all equipped armor and held items with their normal appearance and brightness even when it is partially inside a solid block. Moving the stand downward into the blocks should not cause the stand or its equipment to become dark or black, while the Marker behavior and other intended armor stand properties remain unchanged.

ENVIRONMENT

Minecraft Java Edition Versions: 1.10.2, 16w38a, 16w39a, 16w39b, 16w39c, 16w40a, 16w41a, 16w42a, 16w43a, 16w44a, 1.11 Pre-Release 1, 1.11, 16w50a, 1.11.2, 17w06a, 17w14a, 17w15a, 17w16a, 17w16b, 17w17a, 17w17b, 17w18a, 17w18b, 1.12 Pre-Release 1, 1.12 Pre-Release 2, 1.12 Pre-Release 3, 1.12 Pre-Release 6, 1.12 Pre-Release 7, 1.12, 1.12.1 Pre-Release 1, 1.12.1, 1.12.2 Pre-Release 1, 1.12.2 Pre-Release 2, 1.12.2, 17w43a, 17w43b, 17w45b, 17w46a, 17w47a, 17w47b, 17w48a, 17w49a, 17w49b, 17w50a, 18w01a, 18w03b, 18w05a, 18w16a, 18w22a, 1.13-pre1, 1.13-pre2, 1.13-pre3, 1.13-pre4, 1.13-pre5, 1.13-pre6, 1.13-pre7, 1.13-pre8, 1.13,

18w31a, 18w33a, 1.13.1-pre1, 1.13.1, 1.13.2-pre2, 1.13.2, 18w45a, 18w46a, 18w47b, 18w48a, 18w48b, 18w49a, 18w50a, 19w02a, 19w03b, 19w03c, 19w04b, 19w05a, 19w06a, 19w07a, 19w12b, 19w13b, 1.14.1, 1.14.2 Pre-Release 2, 1.14.2 Pre-Release 3, 1.14.3, 1.15.1, 1.15.2 Pre-Release 1, 1.15.2, 20w06a, 20w09a, 20w12a, 20w13a, 20w13b, 20w14a, 20w17a, 20w19a, 1.16 Pre-release 5, 1.16 Release Candidate 1, 1.16, 1.16.1, 20w28a, 1.16.2 Pre-release 1 (includes snapshots)

ORIGINAL DESCRIPTION

Despite the fact that Armor Stands got Marker:1, they are rendering black because part of their hitbox (the top) is inside a solid block.

Steps to reproduce

1. Place a commandblock
2. Stack 3 solid blocks on top of the CommandBlock, e.g. smoothstone
3. Insert the following command into the CommandBlock and trigger it:

```
/summon armor_stand ~ ~2 ~ {CustomNameVisible:1b,Glowing:1b,ShowArms:1b,Marker:1b,Invisible:0b,NoBasePlate:0b,HandItems:[{id:"minecraft:lapis_block",Count:1b},{id:"minecraft:lapis_ore",Count:1b}],ArmorItems:[{id:"minecraft:iron_boots",Count:1b},{id:"minecraft:iron_leggings",Count:1b},{id:"minecraft:chainmail_chestplate",Count:1b},{id:"minecraft:player_head",Count:1b,tag:{SkullOwner:{id:"994924dc-8404-4af4-a58b-678426ee0095",Properties:{textures:[{Value:"eyJ0ZXh0dXJlcyl6eyJTS0lOlp7InVybCI6Imh0dHA6Ly90ZXh0dXJlc5taW5lY3JhZnQubmV0L3RleHR1cmUvYzBmZmE0OWlxYWQ3ZmNmZDZhNWUxYTU4YzIxMzJfFkM2I5OTYwMDZjMTAzODBhNTQ0MTlkZmFmM2ZiNzc4MzdjZiJ9fX0="}}}],CustomName:"{'text':'Meri','color':'dark_blue','bold':'true','italic':'true'}}
```

4. teleport the Armor Stand gradually down, e.g. with:

```
/execute as @e[type=armor_stand] at @s run tp @s ~~-0.1 ~
```

At some point you'll see that the Armor Stand including its equipment goes dark.

As AS are used a lot not only outside, but also inside of blocks, and using transparent blocks instead to circumvent the AS incl. their rendered equipment going dark is not always a possible option, it'd be desirable for mapmakers to have Marker:1-AS rendering themselves and their equipment "dark" fixed, while keeping the other needed fixes for AS that already were made or that are currently in the making.

But, as already pointed out, it'd be very good for mapmakers if Marker:1 would not render AS/their equipment dark if stuck inside a solid block, as more options would then be possible for us.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-107529>